Pacing - Temporary Transvenous Procedure

1. Scope

Site	Operational Area	Applicable to
Armadale Health Service	Intensive Care Unit	Nursing and Medical

2. Purpose

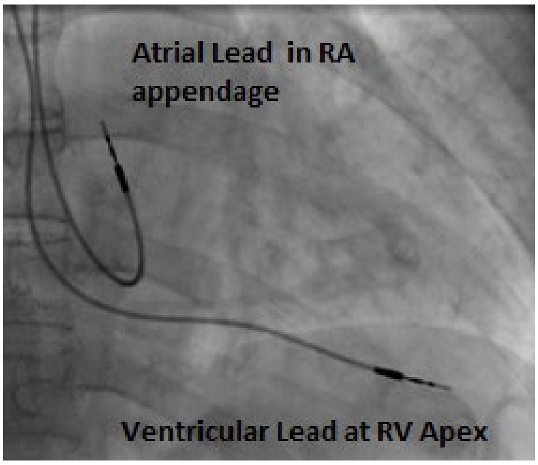
Continuous cardiac monitoring is required for patients with temporary pacemakers – from insertion until removal of the temporary transvenous pacing wire (TVPW).

3. Procedure

Temporary pacing typically serves as a bridge to a more definitive solution to a low heart rate, such as permanent pacemaker implantation, or resolution of a transient or reversible cause.

Temporary transvenous cardiac pacing involves the insertion of a pacing wire, via a vein onto the endocardial wall of the right atrium or right ventricle (*figure 1*) so that a small electrical current can be delivered to the myocardium

Figure 1



3.1 Indications for use

- Syncope
- Slow atrial fibrillation
- Symptomatic second-degree heart block
- Symptomatic complete heart block
- Asystole
- Heart blocks following acute myocardial infarction
- Overdrive pacing of tachyarrhythmia’s
- Drug toxicity or overdose with drugs that can cause bradyarrhythmias

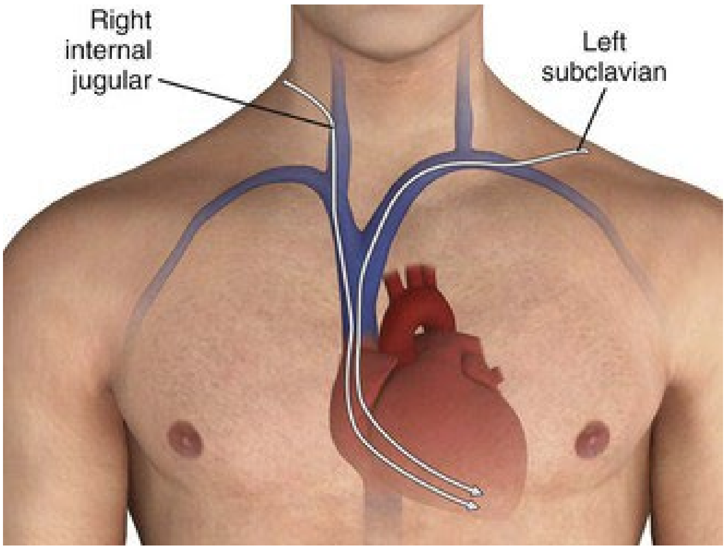
- Pre permanent pacemaker box change for patients with no underlying rhythm
- Pre elective surgical support if patient has history of bradyarrhythmias under anaesthesia with no underlying rhythm
- Prevention of bradycardia in patients with bradycardia induced tachyarrhythmias

3.2 Potential complications of the procedure

- Myocardial perforation causing cardiac tamponade
- Valvular damage-tricuspid regurgitation
- Ventricular/Atrial perforation
- Venous thrombosis
- Pneumothorax
- Haemothorax
- Pulmonary embolism
- Air embolism
- Atrial and ventricular arrhythmias during catheter insertion (e.g. AF, VT, VR or asystole)
- Bleeding
- Nerve damage
- Infection

3.3 Insertion site choice and complications

- **Subclavian** – most reliable but may not be used if permanent pacemaker insertion required as it reduces number of insertion sites
- **Internal jugular** – lower incidence of complications than subclavian route
- **External jugular** (right) – offers the most direct route to the right ventricle, and is associated with the highest success rate and fewest complications
- **Femoral vein** – route of choice post thrombolysis. Higher incidence of infection when femoral vein used. Patient mobility is limited with TVPW in situ
- **Brachial vein** – Safe route but can be technically difficult. Can be used post-thrombolysis



The right internal jugular and left subclavian veins have the straightest anatomic pathway to the right ventricle and are generally preferred for temporary transvenous pacing.

3.4 Post-insertion complications

- Lead displacement or fracture causing intermittent or complete failure to capture and/or sensing issues.

3.5 Required Equipment

- Contact Radiology to provide fluoroscopy – Image intensifier – C Arm – Kept in theatre
- ICU beds allow for fluoroscopy screening
- Resus trolley with Defib / External pacer
- Procedure trolley, ICU packs and appropriate PPE (Sterile gown, Glove, cap etc.)
- A prepared kit with equipment for temporary transvenous pacing wire (TVPW) insertion is located in the ICU Store
- Specific Equipment
- Pulse Generator / Pacing box (Medtronic).
- Bridging cable / connecting cable to pacing box
    - White bridging cable = Ventricular cable
    - Blue bridging cable = Atrial (generally not used)
- Introducer sheaths 6 Fr x 2
- Pacing wires 5 Fr x 2
- Scalpel Blade x 2
- Spare Guide wires x 2
- Cooks (Potts) needle x 3

3.5.1 General

- Syringes 10 ml x 3
- Gauze swabs x 8
- Needles 23 & 26 x 3 each
- Lignocaine 2%
- Chlorhexidine 2% and Alcohol 70% Or Betadine (caution Iodine allergies)
- Suture material curved (2.0) x 2
- Dressing- IV 3000 or CHG Tegaderm
- Fixomull x 1 roll
- Sodium chloride ampules x 5
- Bung for side arm of sheath

3.6 Patient Preparation

- Emergency and life-threatening situations (e.g. intubation) may limit the amount of information that can be provided to the patient and family/significant others and may reduce the amount of time to respond to questions from the patient or family/significant others.
- Explain procedure to patient
- Confirm that patient has patent intravenous (IV) access.
- Position defibrillation trolley with emergency drugs available next to patient's bed/trolley
- Ready access to multifunction electrodes (i.e. transthoracic pacing pads)

3.7 Nursing Actions during TVPW Insertion

- Reassure patient throughout procedure as required
- 5-minutely haemodynamic observations should be performed without compromising patient or staff safety.
- Observe the cardiac monitor while the operator (MO) inserts the pacing wire
- Due to the potential risk of micro shocks when handling the pacing electrode, it is advisable to wear gloves
- Notify the operator of changes in the patient's cardiac rhythm. Assist the operator (MO) to connect TVPW to the pacemaker generator
- Turn the temporary pacemaker ON and adjust pacing settings as instructed.

3.8 Post-TVPW Insertion

- Ensure the pacing wire is secured as it exits the introducer. The operator will suture the TVPW in place.
- Turn monitor pacing alarms on at bedside monitor.
- Loop pacemaker extension lead to prevent tension on TVPW-ensure all connections are easily accessible for trouble shooting
- Perform a 12 lead ECG to evaluate correct function of temporary pacemaker
- Obtain a portable chest X-ray post-TVPW insertion
- Set back up pulse generator/s to the same setting/s as the pulse generator in use (if model of pacemaker generator is able to retain settings)
- The backup pulse generator is left off to preserve battery life and is to remain at the bedside

3.9 Documentation

- Record: By the medical team
- Insertion of TVPW in patient integrated notes and ICU Flow chart
- Pacemaker settings - rate, mode, output, threshold, sensitivity, A-V intervals and underlying / intrinsic cardiac rhythm
- Update if settings change
- Patient mobility, diet and fluid restrictions, TVPW dressing on Patient Care Plan
- Set pacemaker rate, output (mA), and sensitivity (mV - demand or asynchronous), as determined by the threshold testing and prescription.

Daily ECG of underlying rhythm ONLY if the patient is haemodynamically stable

3.10 Determine Pacing Thresholds

Testing and determining the pacing and sensing threshold is performed under the direct supervision of an **Experienced Consultant / Senior Nurse**.

Thresholds should be **checked/tested daily and prn**.

3.10.1 Determine the Stimulation (Pacing) Threshold

- Patient should be returned to bed prior to threshold testing
- Set pacing rate above patient's intrinsic rate (if patient not 100% paced rhythm).

- The patient must be consistently 100% paced to test the pacing threshold
- Gradually decrease output (mA) until capture is lost.
    - The output dial regulates the amount of electrical current (mA) that is delivered to the myocardium to initiate depolarisation
- Patient may experience haemodynamic instability. Loss of capture during testing may result in reduction in cardiac output
- Once capture is lost - Increase output (mA) until capture is re- established. This is the stimulation threshold.
- Ventricular pacing (stimulation threshold) should be established at less than 1mA output whenever possible
- Set output (mA) at least 1.5 to 2 times higher than the stimulation threshold
- The output setting is sometimes referred to as the maintenance threshold

3.10.2 Determine the Sensitivity Threshold

Must only be performed on a patient **with an intrinsic rhythm** and who is haemodynamically **stable.**

- Sensitivity threshold is the level at which intrinsic electrical activity is recognised by sensing electrodes
- Set output to 0.1mV to reduce the risk of ‘R on T’ phenomenon (Initiating a pacing spike on the T wave - may induce ventricular tachycardia)
- Set rate at least 10 beats per minute below patient’s intrinsic heart rate.
- Set sensitivity control to most sensitive setting (fully demand or lowest numerical setting 0.5mV). The Yellow Sensing indicator light should flash with **each intrinsic R wave.**
- Gradually decrease the sensitivity level by turning the sensitivity dial counterclockwise.
    - This Increases the intrinsic mV that must be sensed by the pacemaker – (turning the dial from .5mV towards 20mV until the pacemaker is pacing at all times (Asynchronous)
    - Note the level that the pacemaker becomes asynchronous
    - Turn the dial back towards the lower end of the dial (clockwise) until the yellow “Sensing” light is flashing with each intrinsic R Wave
    - Note this level - this is the sensitivity threshold
- The sensitivity threshold is set at maximum and is lowered only if pacer is sensing inappropriately
    - If sensitivity is set too high (minimal intrinsic electrical current to be sensed by the pacemaker = Low mV), the pacemaker may sense muscle activity and or P or T waves as being R waves and inhibit the pacemaker from firing.
    - If sensitivity is set too low (larger electrical currents must be sensed by the pacemaker = High mV), the pacemaker will not sense the patient’s own cardiac rhythm and results in asynchronous pacing (non-sensing). This increases the risk of ‘R on T’ phenomenon (Initiating a pacing spike on the T wave) which may induce ventricular tachycardia.

3.10.3 A-V Interval

A-V Interval applies to dual chamber pacing only. Dual chamber pacing will not be used at AHS.

3.10.4 Assess Rhythm for Appropriate Pacemaker Function

Capture: Depolarisation of a cardiac chamber in response to a pacing stimulus

- Is there a broad QRS complex for every ventricular pacing spike?

Rate: is the rate at or above the pacemaker rate if in the demand mode?

Sensing: The ability of the pacemaker to detect intrinsic cardiac activity and respond appropriately.

- Does the sensitivity light on the pulse generator indicate that every QRS complex is sensed (should flash with each beat)

The ECG tracing should reflect appropriate response to pacemaker settings if the pacemaker is functioning properly.

- Assess the patient for both electrical and mechanical capture.

Refer to Appendix 1: Pacing Codes and Modes of Pacing

3.10.5 Pacing Examples

Atrial pacing is the delivery of an electrical stimulus to the atrium. Atrial pacing appears on the ECG as a single pacemaker stimulus followed by atrial depolarisation (or a P wave). The morphology of the P wave depends upon the location of the atrial lead; it may be normal, diminutive, biphasic, or negative	
Ventricular pacing is the delivery of an electrical stimulus to the ventricle. Indicated on the (ECG) as a single pacemaker spike prior to ventricular depolarisation, and followed by a QRS complex that is wide, bizarre, and resembles a ventricular beat	
Dual chamber pacing is the delivery of an electrical stimulus to the atrium and/or the ventricle. Atrioventricular (AV) sequential pacing appears on the electrocardiogram as pacemaker spikes before the P wave and QRS complex	

3.11 Troubleshooting pacemakers

Refer to Appendix 2: Troubleshooting

3.12 Removal of the Transvenous Pacing Wire

- TVPW should only be removed on the instruction of the medical ICU team
- Inform MO and Shift Coordinator when removing transvenous pacing wire.
  Equipment
- Basic dressing pack
- PPE as per standard precautions
- Stitch cutter
- Chlorhexidine 1% in isopropyl alcohol 70% solution
- Occlusive dressing

3.13 Removal Procedure

Ensure balloon is deflated on balloon tipped TVPW prior to removal. Removal with balloon inflated may cause valve damage, vascular damage, haemodynamic collapse and death.

1. Turn off pacemaker
2. With clean hands prepare trolley and gather equipment
3. Perform hand hygiene
4. Open sterile dressing pack and equipment
5. Don PPE as per standard precautions
6. Remove dressing and discard
7. Perform hand hygiene
8. Clean insertion site, sheath and surrounding area with chlorhexidine 2% in isopropyl alcohol 70% solution using gauze swabs in concentric circles from inner to outer
9. Cut suture securing pacing electrode (as required) – Do not cut suture securing sheath.
10. Ask patient to perform Valsalva manoeuvre (not required for femoral/brachial inserted wires).
11. With non-dominant hand apply constant, gentle pressure to the sheath whilst removing TVPW with dominant hand

If it is deemed that the TVPW will not require reinsertion; the sheath is to be removed by following the steps 2 to 8 above

- With non-dominant hand apply constant, gentle pressure to the exit site whilst removing the sheath with your dominant hand
- Apply gauze to exit site and maintain direct pressure until haemostasis is achieved
- Cover insertion site with occlusive dressing. Dressing remains sealed for 24 hours.
- Dispose of equipment, waste and gloves
- Clean trolley surface
- Perform hand hygiene
- Document removal in patient integrated notes. Update Patient Care Plan.

4. Legislative context

The following documents are required to give effect to this procedure [i.e. the documents included are mandatory]:

- East Metropolitan Health Service (EMHS) Infection Prevention and Management Policy EMHS: 34
- EMHS Consent Policy EMHS: 5
- Recognition and Response Acute Deterioration

5. Supporting information

The following documents support the implementation of this procedure:

- Armadale Kalamunda Group related procedures
- Guideline documents
- Other supporting information from the WA Health Policy Frameworks

6. Compliance, monitoring and evaluation

Compliance against this procedure will be evaluated through monitoring of related incidents and trends recorded in the Clinical Incident Management System and investigations as required such as Root Cause Analysis.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.3
- Standard 3 – Action 3.10
- Standard 4 – Action 4.3
- Standard 5 – Action 6.5
- Standard 8 – Action 8.4References

1. With Acknowledgement to the Royal Perth Hospital Nursing Practice Standard for Temporary Transvenous Cardiac Pacing Patient Management
2. Timothy, P.R. and Rodeman, B.J. (2004) Temporary Pacemakers in Critically Ill Patients. Assessment and Management Strategies. AACN Clinical Issues, 15(3).

8. Policy owner (relating to these procedures)

Head of Department Intensive Care Unit

Enquiries relating to this procedure may be directed to:

Title: Coordinator of Nursing
Department: Surgical
Email: Shane.Hastings@health.wa.gov.au

9. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V2	30/10/2024	30/10/2027	Scheduled review

10. Authorisation

This procedure has been authorised and issued by Director of Clinical Services.

Authorised by	Dr Tania Strickland – A/Director of Clinical Services
Authorisation date	25/10/2024
Published date	30/10/2024

Appendix 1: Pacing Codes and Modes of Pacing

1st Letter	2nd Letter	3rd Letter
Chamber Paced	Chamber Sensed	Response to Sensing
A = Atrium	A = Atrium	T = Triggered
V = Ventricle	V = Ventricle	I = Inhibited
D = Dual (Atrium & Ventricle)	D = Dual (Atrium & Ventricle)	D = Dual (Triggered & Inhibited
O = None	O = No sensing	O = No response

Adapted from Critical Care Nursing, Diagram and Management 5, 12. Table 2: Modes of Pacing

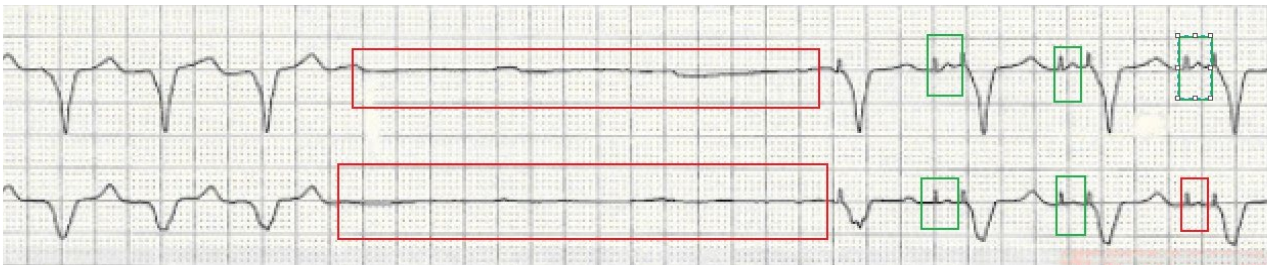
Pacing Mode	Description
Fixed Rate (Asynchronous) AOO VOO DOO	Usually only used in operating theatre or ICU to reduce inappropriate oversensing of diathermy, shivering etc.) Atrial pacing, no sensing Ventricular pacing, no sensing Atrial and ventricular pacing, no sensing
Demand (Synchronous) AAI VVI DVI	Atrial pacing, atrial sensing, inhibited response to sensed P waves Ventricular pacing, ventricular sensing, inhibited response to sensed QRS complexes Atrial and ventricular pacing, ventricular sensing, both atrial and ventricular pacing will be inhibited if ventricular activity sensed
Universal DDD	Atrial and ventricular pacing and sensing, inhibited response to sensed events in respective chamber, triggered ventricular response to sensed atrial activity if ventricular activity not sensed to upper rate level

Appendix 2 - Troubleshooting

The ability to initiate transthoracic (External) pacing should be present at all times for patients with a TVPW in the event of transvenous pacing failure.

Failure to Pace

- Results in disappearance of the pacing spike, even though the patient’s heart rate is lower than the set pacing rate.
- Failure to pace occurs when the pacemaker does not generate an electrical impulse.
- On an ECG tracing, pacemaker spikes will be missing. Pulse rate will be slower than set rate on the pacing box.



Red boxes highlight areas where there was a failure to Pace.

Green areas show normal pacing and capture

Potential Causes

- Disconnection in the pacing circuit
- Flat battery
- pacemaker turned off or programmed incorrectly
- Pacing lead fracture
- Malfunction of pacing box.
- Fibrosis of myocardial tissue
- Lead displacement
- Inappropriate inhibition of pacing
- Electromagnetic interference

Management

- Assess and document patient’s vital signs and heart rhythm
- Activate MET call if patient haemodynamically compromised/unresponsive. Refer to Recognition and Response Acute Deterioration policy.
- Check all connections, leads and wires are intact
- Press emergency asynchronous pacing button or turn sensitivity dial to asynchronous (Pacemaker generator box variants)
- Decrease sensitivity (mV) (increase the number on the sensitivity dial)
- Increase output (mA)
- Check and change battery/pacing box if necessary

- Commence transthoracic pacing if necessary
- Change patient position i.e. put on left side
- Inform MO/Shift Coordinator urgently

Failure to capture

Failure to capture is evident on ECG by the presence of pacing spikes which are not followed by a P wave (Atrial Pacemaker) or a QRS complex (Ventricular Pacemaker).

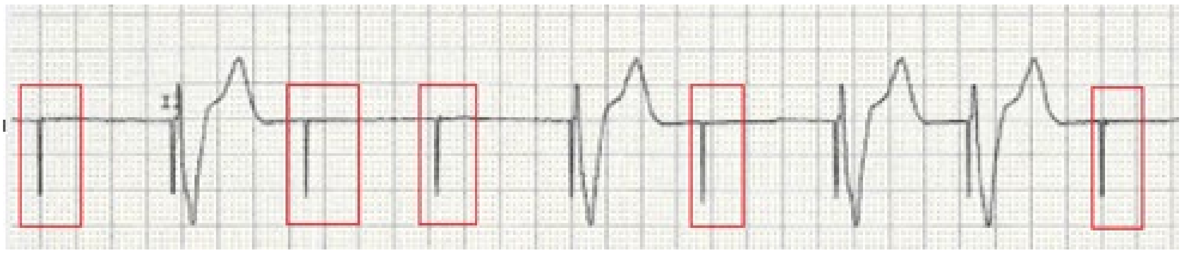


Figure 4:

Potential Causes

- Lead displacement – Common cause
- Failing battery
- Energy output set too low
- Malfunction of pacing box
- Fibrosis / scar tissue of myocardial tissue
- Electrolyte imbalances

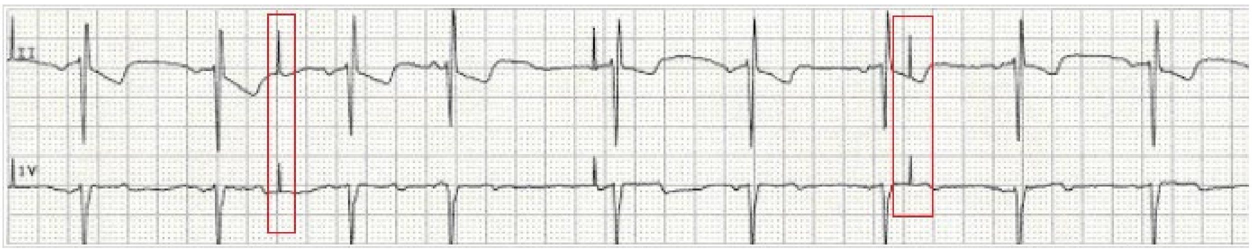
Management

- Assess patient vital signs (heart rate, blood pressure, respiratory rate and oxygen saturation) and heart rhythm
- Increase output (increase the number on the output dial)
- Notify MO and Shift coordinator urgently
- Reposition patient e.g. left lateral if lying on back
- Activate MET or cardiac arrest if patient haemodynamically compromised/unresponsive.
- Refer to Recognition and Response Acute Deterioration policy.
- Commence transthoracic pacing if necessary

Failure to Sense – Undersensing

Failure to sense occurs when the pacemaker fails to detect the hearts intrinsic electrical activity and discharges electrical impulses (paces), thus competing with intrinsic activity.

Pacing spikes may occur at any point in the cardiac cycle and occur unrelated to spontaneous complexes



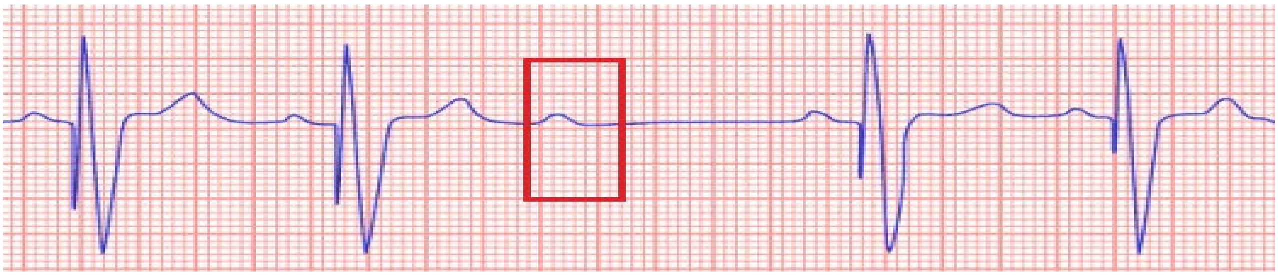
Potential Causes

- Failing battery
- Pacemaker sensitivity set too low
- Malfunction of pacing box
- Lead displacement or fracture
- Fibrosis / scar tissue of myocardial tissue
- Electrolyte imbalances
- Medications

Management

- Assess patient vital signs (heart rate, blood pressure, respiratory rate and oxygen saturation) and heart rhythm
- Increase sensitivity (decrease millivolts). i.e. decrease the number on the dial
- Notify MO and Shift coordinator urgently
- Reposition patient e.g. left lateral if lying on back
- Change battery
- Activate MET or cardiac arrest if patient haemodynamically compromised/unresponsive.
- Refer to Recognition and Response Acute Deterioration policy.
- Commence transthoracic pacing if necessary

Failure to Sense - Oversensing



Oversensing occurs when there is inappropriate sensing of extraneous electrical signals (skeletal muscle activity, electrical interference). These electrical signals are interpreted by the pacemaker as intrinsic activity and as a result the pacemaker does not fire. It is recognized by absent pacemaker spikes and ventricular asystole. These inappropriate signals may be large P or T waves, skeletal muscle activity or **lead contact problems. Abnormal signals may not be evident on ECG.**

Management

- Decrease the sensitivity by increasing the millivolts (increase the number on the dial)
- Turn to asynchronous pacing if ventricular pacing is being inhibited and patient is haemodynamically compromised